

IMMUNE SYSTEM

THE COMPLEXITIES OF THE IMMUNE SYSTEM CREATE THE CONDITION OF IMMUNITY, IN WHICH, AFTER THE FIRST ATTACK BY A PARTICULAR TYPE OF MICROORGANISM, THE BODY IS PROTECTED OR RESISTANT TO FUTURE INVASIONS.

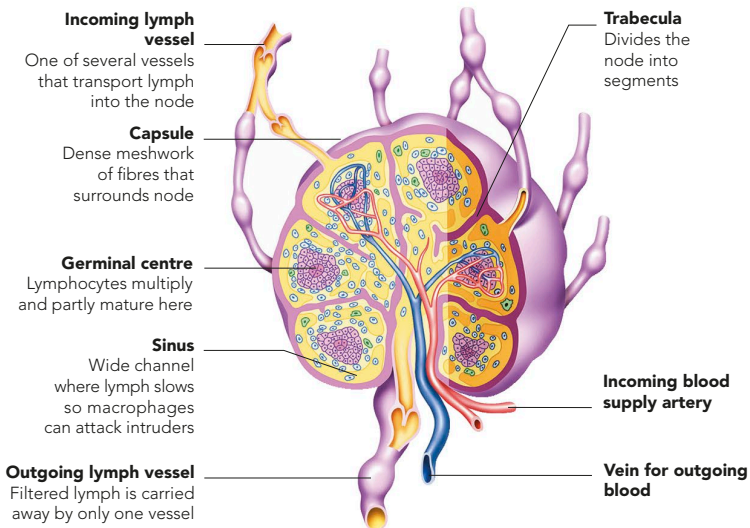
LYMPH NODES

The lymph nodes produce and harbour lymphocytes that protect the body from disease. They are scattered throughout the body and are also concentrated in groups (see p.192). Small lymphatics

(vessels) bring lymph to a node, while a larger vessel carries it away. The nodes filter and clean the lymph, which then drains into the venous bloodstream. Lymph vessels have valves so the fluid flows one way.

INSIDE A NODE

A lymph node, or gland, is a mass of segmented lymphatic tissue covered in a fibrous capsule. It contains sinuses, where many scavenging white blood cells, called macrophages, ingest bacteria as well as other foreign matter. Lymph nodes vary in diameter from 1 to 25mm (1/25 to 1in), although they can swell during infection or illness.



WHITE CELL TYPES

There are numerous types of white blood cell, which are known by the general name of leucocytes. All white blood cells are derived from the bone marrow. Some of them grow and mature into other types.

Monocyte

Has a nucleus that is big and rounded, or indented; engulfs pathogens.



The largest leucocytes, and the largest cells in the blood, are monocytes. The lymphocytes are the chief immune cells, and they can be either B- or T- cells, depending on the way the lymphocyte develops.

Lymphocyte

Both types (B- and T- cells) have a large nucleus that almost fills the cell.



Neutrophil

Granulocyte with many particles and multi-lobed nucleus; engulfs pathogens.



Basophil

Granulocyte with lobed nucleus; involved in allergic reactions.



Eosinophil

Granulocyte with B-shaped nucleus; destroys antigen-antibody complexes.

